Big Data Analytics

UNIT 1: INTRODUCTION

What is data? What are the characteristics of data?

Data - Data is nothing but individual fact(s). Data can be numbers, words, measurements, observations or just descriptions of things.

Data is a collection of information gathered by observations, measurements, research or analysis. They may consist of facts, numbers, names, figures or even description of things.

Data refers to the quantities, characters, or symbols on which operations are performed by a computer, which may be stored and transmitted in the form of electrical signals and recorded on magnetic, optical, or mechanical recording media.

Examples: price of a movie ticket in INR, weight of tomatoes in kilograms, address of your friend, age, name, temperature in centigrade, date, distance in kilometres, and so on.

Data can be either quantitative or qualitative.

Quantitative Data - It is quantifiable and in numerical form. Examples include the weight, volume, or cost of an item, or age of a person, or distance between two cities.

Qualitative Data - It is descriptive, but non-numerical, like the name, gender, or eye colour of a person, or mood (Happy, Satisfied, Sleepy, Angry, Frustrated, Sad, …)

Characteristics of Data - Data is usable and valuable only when it satisfies the following characteristics.

1. Accuracy - Is the information correct in every detail?
2. Validity – Is it valid and legitimate?
3. Reliability - Does the information contradict other trusted resources?
4. Timeliness - How up- to-date is the information? Can it be used for real-time reporting?
5. Relevance - Do you really need this information? Is it relevant to anybody?
6. Completeness - How comprehensive is the information?

Accuracy - Data should be accurate for the intended use and should be captured only once, although it may have multiple uses. Data should be captured at the source where it is generated or at the point of activity.

Validity – Data should be valid or legitimate. Data should be recorded and used in compliance with relevant requirements, rules or definitions. This will ensure consistency with reference to time periods and other factors to reflect whether the data is valid and it measures what is intended to be measured. Example: Ramesh scored 70/100 in Maths when he was in 10th standard of Maharashtra State Board, in the final exam in the year 2022.

Reliability - Data should be reliable. It should reflect stable and consistent data collection processes across collection points and over time. Example: When it comes to election data such as number of constituencies won by a specific political party, the data is reliable when it is collected from the EVMs and reported by the right government authority.

Timeliness - Data should be captured at the right time - as quickly as possible after the event or activity, and must be available for the intended use within a reasonable time period. Data must be available quickly and frequently enough to support information needs and to influence service or management decisions.

Relevance - Data captured should be relevant to the purposes for which it is to be used. This will require a periodic review of requirements to reflect changing needs.

Completeness - Data requirements should be clearly specified based on the information needs of the organization and data collection processes matched to these requirements.

What are the different types of data? Explain with examples. What are the advantages and disadvantages of each type?

Types of Data

There are three types of data.

1) Structured Data
2) Unstructured Data
3) Semi-structured Data

Structured Data: Structured data refers to any data that resides in a fixed field within a record or file. It fits in or complies with a specific Data Model. Structured data is arranged in rows and columns. Structured data has the advantage of being easily entered, stored, queried and analysed.

Examples: Tables in Relational Data Base, spread sheets, Comma Separated Values (CSV)

Structured data is often managed using Structured Query Language (SQL). Implementation of relationships (with Primary Keys, Foreign Keys) and indexing enables fast access or data retrieval.

Source of Structured Data

Structured data is generated in several systems. These include,

- Online Transaction Processing (OLTP) Systems – Data stored in RDBMS or SQL databases
- Online forms - Data stored in RDBMS or SQL databases
- Manual Data Capturing - Data collected and stored in spreadsheets such as Microsoft Excel
- IoT - Sensors such as GPS or RFID tags – Data stored as Comma Separated Values or data files
- Network and Web server logs
- Medical devices

Advantages of Structured Data

Easy to understand and use: Structured data has a well-defined schema or data model, making it easy to understand and use. This allows for easy data retrieval, analysis, and reporting.

Consistency: The well-defined structure of structured data ensures consistency and accuracy in the data, making it easier to compare and analyse data across different sources.

Efficient storage and retrieval: Structured data is typically stored in relational databases, which are designed to efficiently store and retrieve large amounts of data. This makes it easy to access and process data quickly.

Enhanced data security: Structured data can be more easily secured than unstructured or semi-structured data, as access to the data can be controlled through database security protocols.

Clear data lineage: Structured data typically has a clear lineage or history, making it easy to track changes and ensure data quality. (Data lineage is the process of tracking the flow of data over time, providing a clear understanding of where the data originated, how it has changed, and its ultimate destination.) – For example, from an Employee Management System data that stores data in a set of relational tables, we can find out when a particular employee joined the organization, and when the employee was promoted to different levels and the current designation or status of the employee.

Disadvantages of Structured Data

Rigidity or Inflexibility: Structured data comes with a fixed structure. If we want to accommodate new types of data, we need to make changes to the schema or the data model and it might require significant changes to the database.

Difficult to Model Complex Scenarios: Structured data is often limited in terms of the complexity of relationships between data entities. This can make it difficult to model complex real-world scenarios.

Limited Context: Structured data often lacks the additional context and information that unstructured or semi-structured data can provide, making it more difficult to understand the meaning and significance of the data.

High Cost: Structured data requires the use of relational databases and related technologies, which can be expensive to implement and maintain.

Data Quality: The structured nature of the data can sometimes lead to missing or incomplete data, or data that does not fit cleanly into the defined schema, leading to data quality issues.

Structured data is generated in various ways as listed below.

1. **Created:** Data is created in business applications through functionalities such as user registration, student application, loan application, customer surveys and so on.
2. **Provoked:** Many businesses and websites provide people the opportunity to express their views. Every time a customer rates a restaurant, an employee, a purchasing experience or a product they are creating provoked data. Rating sites, such as Yelp, also generate this type of data.
3. **Transacted:** Businesses collect data on every transaction completed, whether the purchase is completed through an online shopping cart or in-store at the cash register. Businesses also collect data on the steps that lead to a purchase online. For example, a customer may click on a banner ad that leads them to the product pages which then spurs a purchase. Transacted data helps us understand exactly what was bought, where it was bought, and when. Matching this type of data with other information, such as weather, can yield even more insights.

4. **Compiled:** Compiled data is giant databases of data collected on every household. Companies like Acxiom or CIBIL collect information on things like credit scores, location, demographics, purchases and registered cars that other agencies (such as banks and insurance companies) can then access for supplemental consumer data.
5. **Experimental:** Experimental data is created when businesses experiment with different marketing pieces and messages to see which are most effective with consumers. You can also look at experimental data as a combination of created and transactional data.

Unstructured Data: Unstructured data cannot fit into a table or spread sheet because it cannot be conformant to any data model. Indexing is not applied for fast data retrieval.

Examples: Photos and graphic images, videos, streaming instrument data, PDF files, Microsoft PowerPoint presentations, blog entries, wikis and word processing documents.

Sources of Unstructured Data

Unstructured data is available from the internet in the form of web pages, images (JPEG, GIF, PNG, etc.), videos. Across all industries unstructured data is generated in the form of memos, reports, word documents and Microsoft PowerPoint presentations.

Advantages of Unstructured Data

Flexibility - Unstructured data is not constrained by a fixed schema or format and it can accommodate different forms of data. It is very flexible due to the absence of schema.

Portability - Data is portable from one computer to another because data is not stored in a specific RDBMS or SQL database.

Scalability – Unstructured data very scalable because it can span across multiple files or file systems. It can deal easily with the heterogeneity of sources. These types of data have a variety of business intelligence and analytics applications.

Disadvantages of Unstructured data

Difficult to Store and Manage - It is difficult to store and manage unstructured data due to lack of schema and structure.

Speed and accuracy of data retrieval - Indexing the data is difficult and error prone due to unclear structure and not having pre-defined attributes. Due to which data retrieval is slow and search results are not very accurate.

Data Security - Ensuring security to data is difficult task. There is no table level or column level or row level security (as in relational data bases).

Unstructured data is generated in various ways as listed below.

1. **Captured:** Captured data is created passively due to a person's behaviour. Every time someone enters a search term on Google that is data that can be captured for future benefit. The GPS info on our smartphones is another example of passive data that can be captured with big data technologies.

2. **User-generated:** User-generated data consists of all of the data individuals are putting on the Internet every day. From tweets, to Facebook posts, to comments on news stories, to videos put up on YouTube, individuals are creating a huge amount of data that businesses can use to better target consumers and get feedback on products.

Semi-structured Data: Semi-structured data is a form of structured data that does not conform with the formal structure of data models associated with relational databases or other forms of data tables. However, semi-structured data contain tags or other markers to separate semantic elements and enforce hierarchies of records and fields within the data. Therefore, it is also known as self-describing structure. It is also called quiz data or data in between structured and semi structured. It is partially structured data but it does not follow a data model. It does not have a rigid structure.

Examples: E-mails, log files, HTML files, XML files, and so on.

Word processing software, XML and other markup language are often used to manage semi structured data.

Semi-structured data, or partially structured data, doesn't follow the tabular structure associated with relational databases or other forms of data tables. However, it does contain tags and metadata to separate semantic elements and establish hierarchies of records and fields.

Semi-structured data comes in a variety of formats, such as:

XML: Extensible Markup Language (XML) has become one of the most popular semi-structured data formats. This versatile and easy-to-use markup language allows users to define tags and attributes required for storing data in a hierarchical form.

JSON: A commonly used alternative to XML, JavaScript Object Notation (JSON) collects semi-structured data from IoT devices, web browsers, and smartphones, then organizes that data into batches before transmitting it to a data platform via a data pipeline. This versatile format can also be used to transfer data in between servers and apps or internet-connected devices.

Avro: Originally developed for use with Apache Hadoop, Avro is a remote procedure call (RPC) framework and data serialization. Using schemas defined in JSON, Avro serializes data in a compact, binary format that can be sent to any app or program where it is deserialized.

ORC: Optimized Row Columnar (ORC) is a semi-structured data format that was initially designed to achieve more-efficient compression and enhance performance for reading, writing, and processing data over earlier Hive formats. (The Apache Hive ™ is a distributed, fault-tolerant data warehouse system that enables analytics at a massive scale and facilitates reading, writing, and managing petabytes of data residing in distributed storage using SQL.)

Sources of Semi-structured Data:

• E-mails • XML and other markup languages • Binary executables • TCP/IP packets • Zipped files • Integration of data from different sources • Web pages

Advantages of Semi-structured Data:

Flexibility - The data is not constrained by a fixed schema. Schema can be easily changed. It is possible to view structured data as semi-structured data. It supports users who cannot express their need in SQL. It can deal easily with the heterogeneity of sources.

Portability – The data is portable across systems.

Data Storage and Management - Semi-structured data provides more flexibility in terms of data storage and management, as it can accommodate data that does not fit into a strict, predefined schema. This makes it easier to incorporate new types of data into an existing database or data processing pipeline.

Scalability - Semi-structured data is particularly well-suited for managing large volumes of data, as it can be stored and processed using distributed computing systems, such as Hadoop or Spark, which can scale to handle massive amounts of data.

Faster data processing - Semi-structured data can be processed more quickly than traditional structured data, as it can be indexed and queried in a more flexible way. This makes it easier to retrieve specific subsets of data for analysis and reporting.

Improved data integration - Semi-structured data can be more easily integrated with other types of data, such as unstructured data, making it easier to combine and analyse data from multiple sources.

Richer data analysis - Semi-structured data often contains more contextual information than traditional structured data, such as metadata or tags. This can provide additional insights and context that can improve the accuracy and relevance of data analysis.

Disadvantages of Semi-structured data

Difficulty in Storage - Lack of fixed, rigid schema make it difficult in storage of the data

Relationship Interpretation - Interpreting the relationship between data is difficult as there is no separation of the schema and the data. Queries are less efficient as compared to structured data.

Complexity - Semi-structured data can be more complex to manage and process than structured data, as it may contain a wide variety of formats, tags, and metadata. This can make it more difficult to develop and maintain data models and processing pipelines.

Assurance of Data Quality - Semi-structured data often lacks the standardization and consistency of structured data, which can make it more difficult to ensure data quality and accuracy. This can also make it harder to compare and analyse data across different sources.

Performance - Processing semi-structured data can be more resource intensive than processing structured data, as it often requires more complex parsing and indexing operations. This can lead to reduced performance and longer processing times.

Limited Set of Tools - While there are many tools and technologies available for working with structured data, there are fewer options for working with semi-structured data. This can make it more challenging to find the right tools and technologies for a particular use case.

Data Security - Semi-structured data can be more difficult to secure than structured data, as it may contain sensitive information in unstructured or less visible parts of the data. This can make it more challenging to identify and protect sensitive information from unauthorized access.

Also, read - https://www.bigdataframework.org/data-types-structured-vs-unstructured-data/

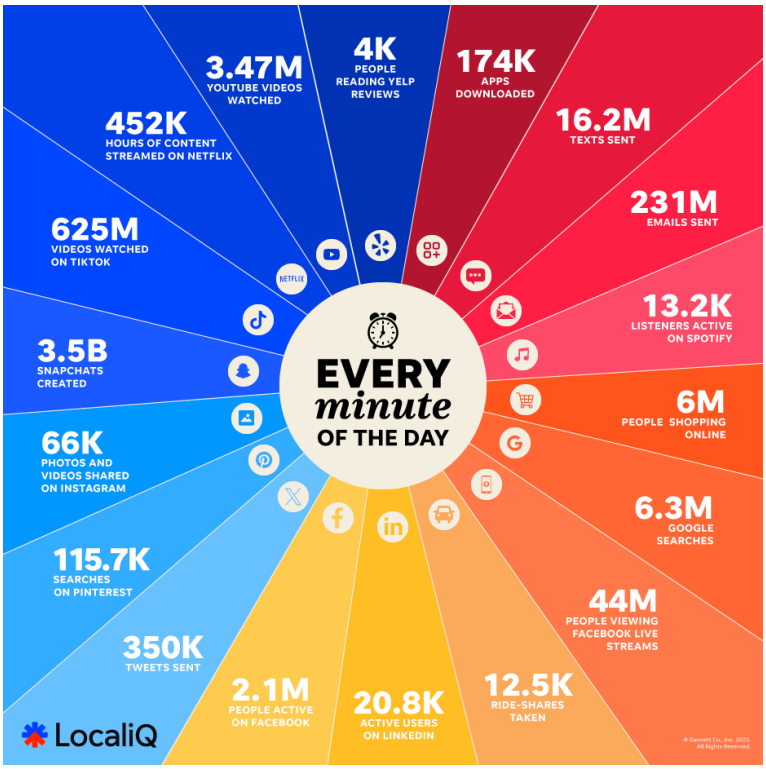
Activity - The various ways in which structured and unstructured data are generated are mentioned in the above pages. Explain how semi-structured data is generated.

1.1 What is Big Data?

Big Data is a collection of data that is huge in volume, yet growing exponentially with time. It is a data with so large size and complexity that none of traditional data management tools can store it or process it efficiently. Here are the key points.

- *Voluminous data that includes structured, unstructured, and semi-structured datasets. It is challenging to manage Big Data using traditional data processing tools. Big Data requires additional infrastructure to govern, analyse, and convert into insights.*
- *Big data is an enormous volume of data that is constantly and rapidly expanding. Traditional data management systems cannot effectively store and analyse this data because of the data is very complex and enormous.*
- *Big data can be mined for information and utilized in advanced applications of analytics like predictive modelling and machine learning.*

Big Data is a field or subject area that deals with approaches, techniques and tools to gather, analyse, systematically extract information and insights from data sets that are too large or complex to be dealt with by traditional data-processing application software.



Online activities happen round the clock around the world. Internet and mobile users are checking emails, sending text messages, buying something for their home, watching Instagram reels, viewing YouTube videos, commenting on blogs, collaborating on discussion forums and performing several online activities.

Online content consumption has increased by 30% in 2023. It is going to increase further in 2024.

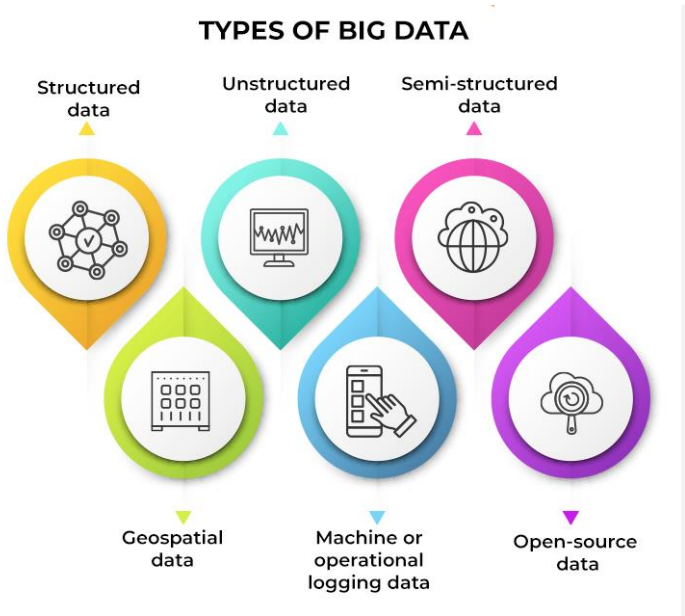
This diagram helps us understand what happens every minute on the internet. From business standpoint, this indicates that customers are busy and distracted. Meanwhile, their online activities are producing massive data.

For more information, read,
https://localiq.com/blog/what-happens-in-an-internet-minute/
https://ediscoverytoday.com/2023/04/20/2023-internet-minute-infographic-by-ediscovery-today-and-ltmg-ediscovery-trends/

1.2 Different types of data generated. Examples of giant companies generating big data.

There are three types of data generated, as discussed in the first few pages of this document.

1) Structured Data
2) Unstructured Data
3) Semi-structured Data



Big data is a voluminous set of structured, unstructured, and semi-structured datasets, which is challenging to manage using traditional data processing tools. It requires additional infrastructure to govern, analyse, and convert into insights.

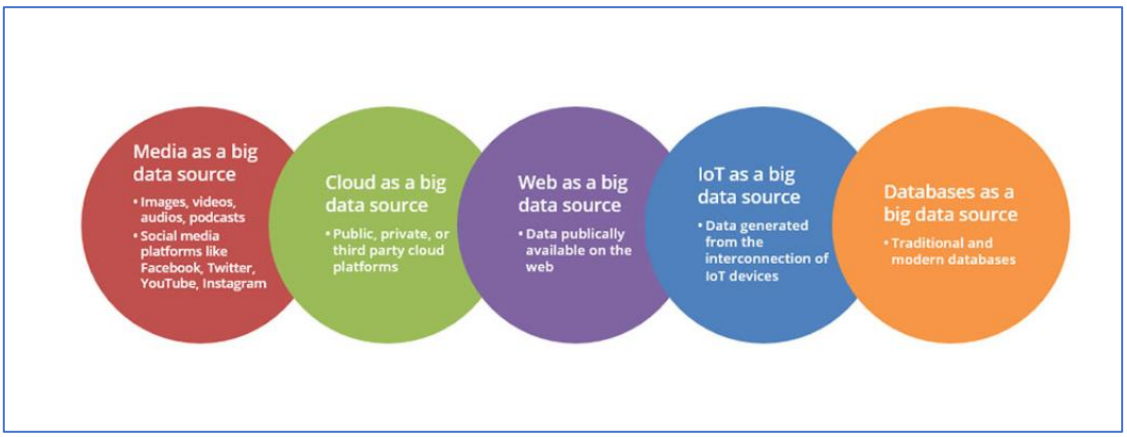
Activity: Read https://www.spiceworks.com/tech/big-data/articles/what-is-big-data/ and understand 'Geospatial data', 'Machine or operational logging data' and 'open-source data'. Write a note on these. Find information about 'Google Public Data Explorer'.

Examples of giant companies generating Big Data

Activity: Search the Internet to identify at least 5 big companies that generate Big Data. For each company find out the size of big data generated (per day or per week or per month). In which areas are they implementing big data analytics? What benefits are they reporting? Visit https://localiq.com/blog/what-happens-in-an-internet-minute/ or similar sites to understand further. In India, which companies are generating big data? Find out.

Sources of Big Data - What are the sources of Big Data? Explain.

Voluminous data is available from various sources. These sources can be categorized as show in the diagram.



Media - Media is the most popular source of big data, as it provides valuable insights on consumer preferences and changing trends. Since it is self-broadcasted and crosses all physical and demographical barriers, it is the fastest way for businesses to get an in-depth overview of their target audience, draw patterns and conclusions, and enhance their decision-making. Media includes social media and interactive platforms, like Google, Facebook, Twitter, YouTube, Instagram, as well as generic media like images, videos, audios, and podcasts that provide quantitative and qualitative insights on every aspect of user interaction.

Cloud – Businesses have invested in cloud infrastructure and shifted their data on the cloud. Cloud storage accommodates structured and unstructured data and provides business with real-time information and on-demand insights. The main attribute of cloud computing is its flexibility and scalability. As big data can be stored and sourced on public or private clouds, via networks and servers, cloud makes for an efficient and economical data source.

World Wide Web - The public web or the Internet constitutes big data that is widespread and easily accessible. Data on the Web or 'Internet' is commonly available to individuals and companies alike. Moreover, web sites such as Wikipedia provide free and quick informational insights to everyone. The enormity of the Web ensures for its diverse usability and is especially beneficial to start-ups and SMEs, as they don't have to wait to develop their own big data infrastructure and repositories before they can leverage big data.

Internet of Things - Data generated from IoT constitute a valuable source of big data. This data is usually generated from the sensors that are connected to electronic devices. The sourcing capacity depends on the ability of the sensors to provide real-time accurate information. IoT is now gaining momentum and includes big data generated, not only from computers and smartphones, but also possibly from every device that can emit data. With IoT, data can now be sourced from medical devices, vehicular processes, video games, meters, cameras, household appliances, and the like.

Databases - Businesses generate and use data with an amalgamation of traditional and modern databases. This results in the accumulation of both structured and unstructured data that is valuable to the business, and becomes a source of big data. This requires a hybrid data model that involves a low investment and IT infrastructural costs. Businesses establish data warehouses, data marts and data lakes for data mining in order to derive value out of corporate data. The extraction of insights from a variety of data are used to grow the business. Popular databases include a variety of data sources, such as MS Access, DB2, Oracle, SQL, and Amazon Simple, among others.

Next, let us understand terms such as datasets, database, data warehouse, data mart, data lake, data analysis and data analytics. This will help us understand Big Data Analytics.

Datasets

Collections or groups of related data are generally referred to as datasets. Each group or dataset member (datum) shares the same set of attributes or properties as others in the same dataset. Some examples of datasets are:

- tweets stored in a flat file
- a collection of image files in a directory
- an extract of rows from a database table stored in a CSV formatted file
- historical weather observations that are stored as XML files

Database

A database is a structured collection of data organized in a way that makes it easy to manage, access, and update. It is a fundamental component of most applications and systems. A database stores all related data related to a specific situation. For example, a point-of-sale (POS) database in a retail store is used to capture and store all the relevant data surrounding the store's transactions.

Databases are single-purpose repositories of raw transactional data and they perform online transactional processing (OLTP).

A database can be a structured, relational database management systems (RDBMS, for example, Oracle or MS Access or MySQL), or unstructured data structures (known as 'NoSQL' for example MongoDB).

Data Warehouse

A data warehouse is a large-scale storage system designed for analysis and reporting. It's optimized for querying and reporting on large volumes of data rather than transaction processing.

Data Mart

A data mart is a subset of a data warehouse. It contains a specific slice of data focused on a particular business function or team within an organization. Data marts are often used to provide specialized views into data for specific departments or groups.

Data Lake

A data lake is a storage repository that can store vast amounts of raw data in its native format. Unlike a data warehouse, which requires data to be structured before storing, a data lake allows you to store data in its raw form and apply structure as needed when performing analysis. It can contain Images, Tables, Files etc.

Analyse the differences between database, data warehouse, data mart and data lake using a tabular format.

	Database	Data Warehouse	Data Mart	Data Lake
Purpose	Used for transactional operations and structured data storage.	Optimized for analytical queries and reporting on structured data.	Created by segmenting the data warehouse into specific subsets.	Made to store large amounts of raw, unstructured or semi-structured data in its raw or unprocessed form for later analysis.
Data Structure	Meant for structured data with predefined schemas.	Works best with structured data but can also handle some semi-structured data.	Scales based on the underlying data warehouse technology.	Mostly for the storage of unstructured, raw, or semi-structured data. Because the data is schema-on-read, you can apply a schema as you query the information.
Use Cases	Applications for managing transactions, structured data, and OLTP (Online Transaction Processing).	Complex querying, analytics, reporting, OLAP (Online Analytical Processing).	Business unit-specific reporting and analytics, focused on a particular area or function.	Handles big data processing, data exploration, machine learning, handling large volumes of raw data.
Scalability	Scales vertically (upgrading resources of a single machine).	Scales horizontally by adding more resources to a cluster.	Scales horizontally by adding more storage.	Scales based on the underlying data warehouse technology.
Cost	Typically, it is more cost-effective for transactional workloads.	Can be more expensive, especially for large-scale analytics workloads.	Cost depends on the scale and complexity of the underlying data warehouse.	Cost-effective for storing huge amounts of raw data, however processing could cost more.
Query Performance	Enhanced for quick, low-latency queries on structured data.	Perfect for complex analytics queries on large datasets.	Performance depends on how it's designed and implemented within the data warehouse.	Due to its schema-on-read design, it can be slower for ad hoc queries but excels at managing enormous amounts of data.

Data Analysis

Data analysis is the process of examining data to find facts, relationships, patterns, insights and/or trends. The overall goal of data analysis is to support better decision-making. A simple data analysis example is the analysis of ice cream sales data in order to determine how the number of ice cream cones sold is related to the daily weather. The results of such an analysis would support decisions

related to the quantity of ice cream a store should order in relation to weather forecast information. Carrying out data analysis helps establish patterns and relationships among the data being analysed.

Data Analytics

Data analytics is a broader term that encompasses data analysis. Data analytics is a discipline that includes the management of the complete data lifecycle, which encompasses collecting, cleansing, organizing, storing, analysing and governing data. The term includes the development of analysis methods, scientific techniques and automated tools. In Big Data environments, data analytics has developed methods that allow data analysis to occur through the use of highly scalable distributed technologies and frameworks that are capable of analysing large volumes of data from different sources.

Big Data Analytics is about gaining a meaningful, deeper, and richer insight into business in order to steer it in the right direction. For example, it helps in understanding the customers' demographics to cross-sell and up- sell products and services to them, and in leveraging the services of vendors and suppliers in order to enhance the business value.

Big Data Analytics is technology-enabled because, several data analytics and visualization tools are available in the market today from leading vendors such as IBM, Tableau, SAS, R Analytics, Statistica, World Programming Systems (WPS), etc.

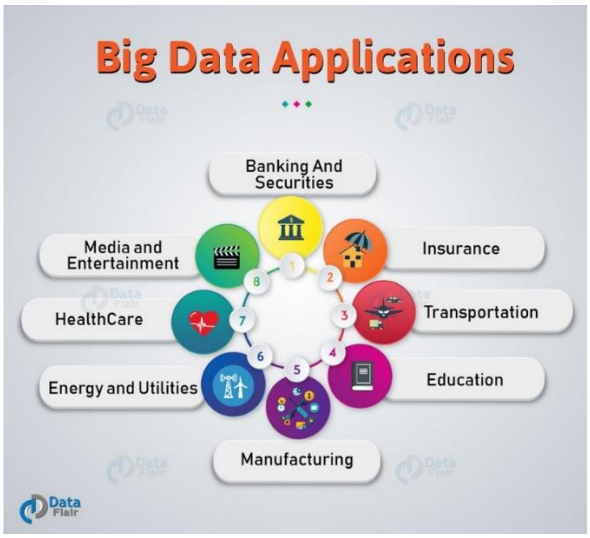
Big Data analytics is the process of examining and interpreting large and complex data sets to identify patterns, trends, and insights that can help senior executives make business decisions, drive innovation, and improve outcomes. It involves the use of advanced data analytics tools and techniques, such as data mining, machine learning, and natural language processing. It can be applied to various domains, such as healthcare, finance, retail, manufacturing, and transportation, among others. Some common use cases of big data analytics include fraud detection, customer segmentation, predictive maintenance, and sentiment analysis.

1.3 Applications of Big Data

Here is a sample list of Big Data applications

1. 360-degree view of the customer
2. Fraud detection and prevention
3. Enhanced personalization and recommendation engines
4. Content analysis of dark data to enhance cyber security and other risks
5. Preventive and proactive maintenance (of devices, appliances, buildings and so on)
6. Customized marketing and discovering new leads (potential customers)
7. Optimization (of fuel, travel route, booking, etc.,)
8. Prediction of user demand (travel and transportation, education, retail products, inventory management…)
9. Observing health conditions via data from wearables
10. Live road mapping or dynamic routing (for drivers)

https://www.geeksforgeeks.org/applications-of-big-data/
https://www.javatpoint.com/applications-of-big-data
https://www.interviewbit.com/blog/applications-of-big-data/
https://www.simplilearn.com/tutorials/big-data-tutorial/big-data-applications



Activity: Do a search on the internet to find details of at least five applications of Big Data.

Sources:
https://www.spiceworks.com/tech/big-data/articles/what-is-big-data/#_004
https://dzone.com/articles/why-is-big-data-in-buzz
https://www.smartdatacollective.com/big-data-20-free-big-data-sources-everyone-should-know/
https://www.smartdatacollective.com/big-data-and-fords-faster-horses/
https://www.allerin.com/blog/top-5-sources-of-big-data
https://www.techtarget.com/searchbusinessanalytics/feature/8-big-data-use-cases-for-businesses-and-industry-examples

1.4 What is Data Science?

Data science is the professional field that deals with turning data into value such as new insights or predictive models. It brings together expertise from fields including statistics, mathematics, computer science, communication as well as domain expertise such as business knowledge. Data scientist has recently been voted the No 1 job in the U.S., based on current demand and salary and career opportunities.

Data Science is the combination of statistics, mathematics, programming, problem-solving, capturing data in ingenious ways, the ability to look at things differently, and the activity of cleansing, preparing, and aligning the data. In simple terms, it is the umbrella of techniques used when trying to extract insights and information from data.

Applications of Data Science – Examples

1) Search engines make use of data science algorithms to deliver the best results for search queries in a fraction of seconds.
2) The entire digital marketing spectrum uses the data science algorithms - from display banners to digital billboards. The goal is to higher Click-Through Rate (CTR).
3) Recommender systems not only make it easy to find relevant products from billions of products available but also adds a lot to user-experience. A lot of companies use recommender system to promote their products and suggestions in accordance with the user's demands and relevance of information. The recommendations are based on the user's previous search results.

History of Big Data

Although the concept of big data itself is relatively new, the origins of large data sets go back to the 1960s and '70s when the world of data was just getting started with the first data centres and the development of the relational database.

Around 2005, people began to realize just how much data users generated through Facebook, YouTube, and other online services. Hadoop (an open-source framework created specifically to store and analyse big data sets) was developed that same year. NoSQL also began to gain popularity during this time.

The development of open-source frameworks, such as Hadoop (and more recently, Spark) was essential for the growth of big data because they make big data easier to work with and cheaper to store. In the years since then, the volume of big data has skyrocketed. Users are still generating huge amounts of data—but it's not just humans who are doing it.

With the advent of the Internet of Things (IoT), more objects and devices are connected to the internet, gathering data on customer usage patterns and product performance. The emergence of machine learning has produced still more data.

While big data has come far, its usefulness is only just beginning. Cloud computing has expanded big data possibilities even further. The cloud offers truly elastic scalability, where developers can simply spin up ad-hoc clusters to test a subset of data.

DIKW (Data – Information - Knowledge – Wisdom) Pyramid

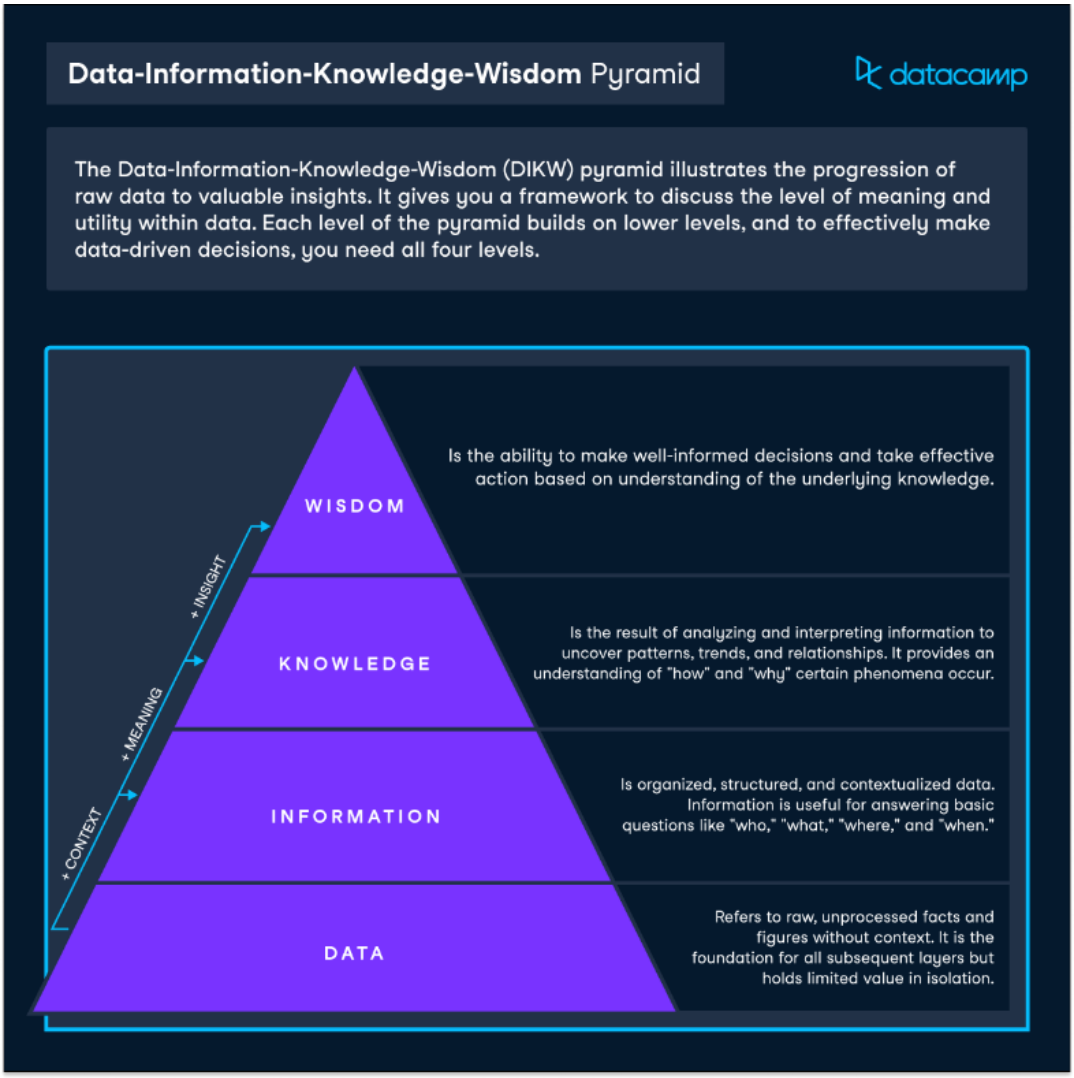
The Data-Information-Knowledge-Wisdom (DIKW) pyramid illustrates the progression of raw data to valuable insights. It gives you a framework to discuss the level of meaning and utility within data. Each level of the pyramid builds on lower levels, and to effectively make data-driven decisions, you need all four levels.

Wisdom is the ability to make well-informed decisions and take effective action based on understanding of the underlying knowledge.

Knowledge is the result of analysing and interpreting information to uncover patterns, trends, and relationships. It provides an understanding of "how" and "why" certain phenomena occur.

Information is organized, structured, and contextualized data. Information is useful for answering basic questions like "who," "what," "where," and "when."

Data refers to raw, unprocessed facts and figures without context. It is the foundation for all subsequent layers but holds limited value in isolation.



Everyday Life Example: Fitness tracking

Fitness tracking devices collect your health and activity data, but your end goal is to use that to make decisions about how to train or how to manage your health.

Wisdom: Understanding these patterns lets you make informed decisions about adjusting your exercise routine, sleep habits, and other lifestyle factors to improve your health and fitness.

⇧

Knowledge: Analysing and interpreting the information may reveal patterns, such as increased step count leading to improved sleep quality or a correlation between heart rate and workout intensity.

⇧

Information: The smartwatch app organizes and structures the data, displaying it in a comprehensible format, such as daily step count, average heart rate, and hours of sleep per night.

⇧

Data: The smartwatch collects raw data such as the number of steps taken, heart rate, and sleep duration.

More Info: https://www.datacamp.com/cheat-sheet/the-data-information-knowledge-wisdom-pyramid

Evolution of Big Data

The first Big Data challenge came in the form of the 1880 U.S. census, when the information concerning approximately 50 million people had to be gathered, classified, and reported on.

With the 1880 census, just counting people was not enough information for the U.S. government to work with—particular elements, such as age, sex, occupation, education level, and even the "number of insane people in household," had to be accounted for. That information had intrinsic value to the process, but only if it could be tallied, tabulated, analysed, and presented. New methods of relating the data to other data collected such as associating occupations with geographic areas, birth rates with education levels, and countries of origin with skill sets, came into existence.

The 1880 census truly yielded a mountain of data to deal with, yet only severely limited technology was available to do any of the analytics. The problem of Big Data could not be solved for the 1880 census, so it took over seven years to manually tabulate and report on the data. With the 1890 census, things began to change, ...

Data Consistency – Soft State and Eventual Consistency - How is data consistency assured in RDBMS and Big Data Management?

RDBMS maintains **ACID** property (Atomicity, Consistency, Isolation, Durability)

Atomicity – Every transaction is atomic. It is completed or not completed. The tasks within a transaction are performed or none of them are. This is the all-or-none principle. If one element of a transaction fails the entire transaction fails.

Consistency - The database must remain in a consistent state at the beginning and end of a transaction; there are never any half-completed transactions. Data needs to be consistent across all related entities.

Isolation – Transactions are isolated from each other. No transaction has access to any other transaction that is in an intermediate or unfinished state. Thus, each transaction is independent unto itself.

Durability - Once the transaction is complete, it will persist as complete and cannot be undone. It will survive system failure, power loss and other types of system breakdowns.

In the world of Big Data, we observe **BASE** (Basically Available, Soft State, Eventual Consistent)

Basically Available – 'Basically Available' means that the system guarantees data availability with reference to CAP Theorem.

CAP theorem states that any distributed system or data store can simultaneously provide only two of three guarantees: consistency, availability, and partition tolerance (CAP). Partition tolerance means that the cluster must continue to work despite any number of communication breakdowns between nodes in the system.

'Basically Available' means that there will be a response to any request. But that response could still be 'failure' to obtain the requested data or the data may be in an inconsistent or changing state, much like waiting for a check to clear in your bank account.

Eventual Consistency – Eventual consistency means that the system will eventually become consistent once it stops receiving input. This is because Big Data keeps growing. It is captured at different intervals. Sooner or later a specific set of data will become consistent. Hence, when a system continues to receive input, it is not checking the consistency of every transaction before it moves onto the next one. (Werner Vogel's article "Eventually Consistent – Revisited" covers this topic is much greater detail.)

Soft State - The state of the system could change over time. Hence, even during times without input there may be changes going on due to 'eventual consistency,' thus the state of the system is always 'soft.'

Big Data – Challenges - What are the challenges in managing Big Data?

High Velocity or Speed of Data Growth - Data growing at a very high velocity poses challenges in processing and extracting insights in a timely manner. There is more and more data generated every

second from which the data that is actually relevant and useful has to be picked up for further analysis.

High Data Volume and Storage Requirements - For organizations, it is challenging to store and manage big data without appropriate tools and technologies. As data sets are becoming bigger and more diverse, there is a big challenge to incorporate them into an analytical platform. If this is overlooked, it will create gaps and lead to wrong messages and insights.

Data Extraction and Synchronization Across Data Sources - When organisations extract or import data from different sources, the data from one source might not be up-to-date as compared to the data from another source. Hence, data extraction and synchronization across data sources is a big challenge.

Data Security - Huge amount of data in organisations can easily become a target for advanced persistent threats, so here lies another challenge for organisations to keep their data secure by proper authentication, data encryption, etc.

Data Reliability (or Veracity) - Big data cannot be 100 percent accurate because it might include redundant or incomplete data, along with conflicts.

Shortage of Talent - There is a huge demand for big data scientists and Big Data analysts in the market. It is important for business organizations to hire data scientists who have adequate skills. Organizations are facing the shortage of professionals who understand Big Data analysis. There is a sharp shortage of data scientists in comparison to the massive amount of data being produced.

Getting Meaningful Insights using Big Data Analytics - It is imperative for business organizations to gain important insights from Big Data analytics, and also it is important that only the relevant department has access to this information. A big challenge faced by the companies in the Big Data analytics is mending this wide gap in an effective manner.

Emerging Tools and Platforms for Big Data Management - With the rise of Big Data, new technologies, tools and platforms come in to existence. A big challenge faced by organizations is to identify tools and platforms that are best suited to them without any potential risks.

Other Challenges – Additional challenges include, integration of data, solution expenses and processing a large amount of data in a timely manner with accuracy so that the data is available for data consumers whenever they need it.

Examples of Big Data

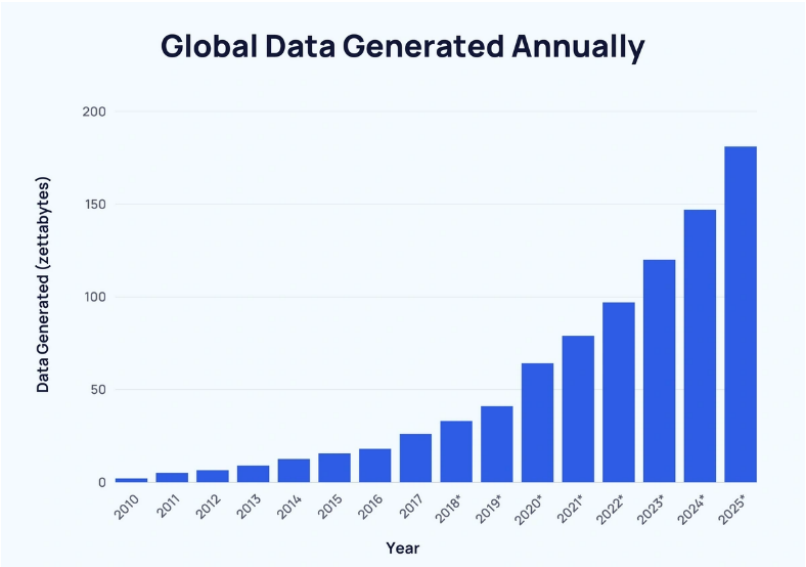
Example 1: The New York Stock Exchange generates about one terabyte of new trade data per day. The statistic shows that 500+terabytes of new data get ingested into the databases of social media site Facebook, every day. This data is mainly generated in terms of photo and video uploads, message exchanges, putting comments etc. A single Jet engine can generate 10+terabytes of data in 30 minutes of flight time. With many thousand flights per day, data grows to many Petabytes.

Activity: Find two more examples of Big Data.

UNIT	SIZE	MORE INFORMATION
Kilobytes (KB)	1000 Bytes	This is the size required by a paragraph in a text document having ~1000 characters.
Megabytes (MB)	1000 KB	This is the size required to store a novel.
Gigabytes (GB)	1000 MB	This is the size required to record and store a concert (music or video).
Terabytes (TB)	1000 GB	This is the storage required to store all the X Rays in a large hospital.
Petabytes (PB)	1000 TB	This is the space required to store half the contents of all US academic research libraries. Internet Giants like Google and Facebook store more than over 100 PBs of data on their data servers.
Exabytes (EB)	1000 PB	About one fifth of the words people have ever spoken. Several hundred exabytes of data are transferred over the Internet every year.
Zettabytes (ZB)	1000 EB	Information in proportion to the grains of sand on all the world's beaches. One zettabyte contains one billion TBs. It will take one billion 1TB hard drives to store one zettabyte of data. Generally, Zettabyte is used to measure large amounts of data and all the data in the world is just a few zettabytes.
Yottabytes (YB)	1000 ZB	It is the largest SI unit of memory measurement.

Activity: Explain 'Data Analytics Life Cycle' with a diagram.

Growth in Big Data (2024)



- Approximately 402.74 million terabytes (or 402.74 billion GB) of data are created each day
- Around 147 zettabytes of data will be generated this year
- 181 zettabytes of data will be generated in 2025
- Videos account for over half of internet data traffic
- The US has over 2,700 data centres

Source: https://explodingtopics.com/blog/data-generated-per-day

Big Data Sources – Examples

Sr. No.	Site	Info
1	http://data.gov/	The US Government pledged in 2022 to make all government data available freely online. This site is the first stage and acts as a portal to all sorts of amazing information on everything from climate to crime.
2	http://www.census.gov/data.html	A wealth of information on the lives of US citizens covering population data, geographic data and education.
3	http://data.gov.uk/	Data from the UK Government, including the British National Bibliography – metadata on all UK books and publications since 1950.
4	https://www.cia.gov/library/publications/the-world-factbook/	Information on history, population, economy, government, infrastructure and military of 267 countries.
5	https://www.healthdata.gov/	125 years of US healthcare data including claim-level Medicare data, epidemiology and population statistics.
6	http://www.hscic.gov.uk/home	Health data sets from the UK National Health Service.
7	http://aws.amazon.com/datasets	Huge resource of public data, including the 1000 Genome Project, an attempt to build the most comprehensive database of human genetic information and NASA's database of satellite imagery of Earth.
8	http://www.google.com/trends/explore	Statistics on search volume (as a proportion of total search) for any given term, since 2004.
9	https://www.google.com/finance	40 years' worth of stock market data, updated in real time.
10	http://wiki.dbpedia.org/	Wikipedia is comprised of millions of pieces of data, structured and unstructured on every subject under the sun. DBPedia is an ambitious project to catalogue and create a public, freely distributable database allowing anyone to analyse this data.

Benefits of Big Data Analytics

Here is a sample list of benefits offered by Big Data Analytics.

1. Identifying the root causes of failures and issues in real time
2. Fully understanding the potential of data-driven marketing
3. Generating customer offers based on their buying habits
4. Improving customer engagement and increasing customer loyalty
5. Reevaluating risk portfolios quickly
6. Personalizing the customer experiences
7. Adding value to online and offline customer interactions

Activity: Explain any 5 benefits of Big Data Analytics in detail.